Thus, there will be sixteen minima between the two first order diffraction minima.

The angular separation between two interference maxima is approximately given by (see Eq. 18.47)

$$\Delta\theta \simeq \frac{\lambda}{d} = 0.904 \times 10^{-4}$$

Thus, the fringe width will be

$$15 \times 12 \times 2.54 \times 0.904 \times 10^{-4} \simeq 0.0413 \text{ cm}$$

18.7 N-SLIT FRAUNHOFER DIFFRACTION PATTERN

LO 6

We next consider the diffraction pattern produced by N parallel slits, each of width b; the distance between two consecutive slits is assumed to be d.

As before, we assume that each slit consists of n equally spaced point sources with spacing Δ (see Fig. 18.32). Thus, the field at an arbitrary point P will essentially be a sum of N terms:

$$E = A\frac{\sin\beta}{\beta}\cos(\omega t - \beta) + A\frac{\sin\beta}{\beta}\cos(\omega t - \beta - \Phi_1) + \ldots + A\frac{\sin\beta}{\beta}\cos(\omega t - \beta - (N-1)\Phi_1) \quad \textbf{(18.48)}$$

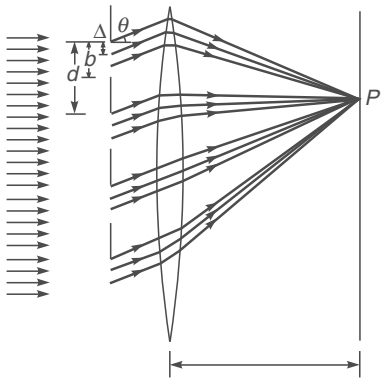


Fig. 18.32 Fraunhofer diffraction of a plane wave incident normally on a multiple slit.

where the first term represents the amplitude produced by the first slit, the second term by the second slit, etc., and the various symbols have the same meaning as in Sec. 18.5. Rewriting Eq. (18.48) we get

$$E = \frac{A\sin\beta}{\beta}\left[\cos(\omega t - \beta) + \cos(\omega t - \beta + \Phi_1) + \ldots + \cos(\omega t - \beta - (N-1)\Phi_1)\right]$$

$$= \frac{A\sin\beta}{\beta}\frac{\sin N\gamma}{\sin\gamma}\cos\left[\omega t - \beta - \frac{1}{2}(N-1)\Phi_1\right] \quad \textbf{(18.49)}$$

where

$$\gamma = \frac{\Phi_1}{2} = \frac{\pi}{\lambda} d\sin\theta$$

The corresponding intensity distribution will be

$$I = I_0\frac{\sin^2\beta}{\beta^2}\frac{\sin^2 N\gamma}{\sin^2\gamma} \quad \textbf{(18.50)}$$

where $I_0 \sin^2\beta/\beta^2$ represents the intensity distribution produced by a single slit. As can be seen, the intensity distribution is a product of two terms; the first term $\left(\frac{\sin^2\beta}{\beta^2}\right)$ represents the diffraction pattern produced by a single slit and the second term $\left(\frac{\sin^2 N\gamma}{\sin^2\gamma}\right)$ represents the interference pattern produced by N equally spaced point sources. For $N = 1$, Eq. (18.50) reduces to the single-slit diffraction pattern [see Eq. (18.10)] and for $N = 2$, to the double slit diffraction pattern [see Eq. (18.45)]. In Figs. 18.33 and 18.34, we have given a plot of the function

$$\frac{\sin^2 N\gamma}{\sin^2\gamma}$$

as a function of γ for $N = 5$ and $N = 12$. One can immediately see that as the value of N becomes very large, the above function would become very sharply peaked at $\gamma = 0, \pi, 2\pi, \ldots$ Between the two peaks, the function vanishes when

$$\gamma = \frac{p\pi}{N}; \quad p = \pm 1, \pm 2, \ldots \quad \text{but} \quad p \neq 0, \pm N, \pm 2N$$

which are referred to as secondary minima.

18.7.1 Positions of Maxima and Minima

When the value of N is very large, one obtains intense maxima at $\gamma \simeq m\pi$, i.e., when

$$d\sin\theta = m\lambda \quad (m = 0, 1, 2, \ldots) \quad \textbf{(18.51)}$$

This can be easily seen by noting that

$$\underset{\gamma\to m\pi}{Lt}\frac{\sin N\gamma}{\sin\gamma} = \underset{\gamma\to m\pi}{Lt}\frac{N\cos N\gamma}{\cos\gamma} = \pm N;$$

thus, the resultant amplitude and the corresponding intensity distributions are given by

$$E = N\frac{A\sin\beta}{\beta} \quad \textbf{(18.52)}$$

and

$$I = N^2 I_0\frac{\sin^2\beta}{\beta^2} \quad \textbf{(18.53)}$$

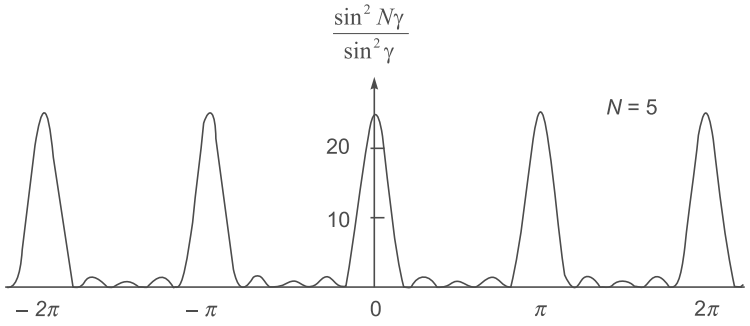


Fig. 18.33 The variation of the function $\sin^2(N\gamma)/\sin^2\gamma$ with γ for $N = 5$.

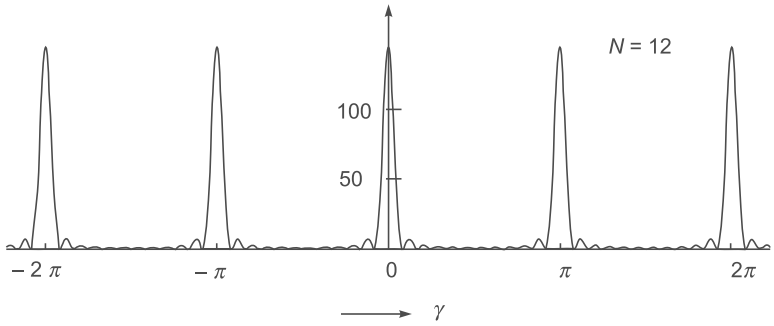


Fig. 18.34 The variation of the function $\sin^2(N\gamma)/\sin^2\gamma$ with γ for $N = 12$. As N becomes larger, the function would become more and more sharply peaked at $\gamma = 0, \pm\pi, \pm 2\pi, \pm 3\pi, \ldots$.

where

$$\beta = \frac{\pi b \sin\theta}{\lambda} = \frac{\pi b}{\lambda}\frac{m\lambda}{d} = \frac{\pi b m}{d} \qquad \textbf{(18.54)}$$

Such maxima are known as **principal maxima**. Physically, at these maxima the fields produced by each of the slits are in phase and, therefore, they add up and the resultant field is N times the field produced by each of the slits; consequently, the intensity has a large value unless $\frac{\sin^2\beta}{\beta^2}$ itself is very small. Since $|\sin\theta| \leq 1$, m cannot be greater than d/λ [see Eq. (18.51)]; thus, there will only be a finite number of principal maxima.

From Eq. (18.50) it can easily be seen that the intensity is zero when either

$$b \sin\theta = n\lambda,\ n = 1, 2, 3, \ldots \qquad \textbf{(18.55)}$$

or

$$N\gamma = p\pi,\ p \neq N, 2N, \ldots \qquad \textbf{(18.56)}$$

Equation (18.55) gives us the minima corresponding to the single slit diffraction pattern. The angles of diffraction corresponding to Eq. (18.56) are

$$d \sin\theta = \frac{\lambda}{N}, \frac{2\lambda}{N}, \ldots, \frac{(N-1)\lambda}{N}, \frac{(N+1)\lambda}{N}, \frac{(N+2)\lambda}{N},$$

$$\ldots, \frac{(2N-1)\lambda}{N}, \frac{(2N+1)\lambda}{N}, \frac{(2N+2)\lambda}{N}, \ldots \qquad \textbf{(18.57)}$$

Thus, between two principal maxima we have $(N-1)$ minima. Between two such consecutive minima the intensity has to have a maximum; these maxima are known as secondary maxima. Typical diffraction patterns for $N = 1$, 2, 3, and 4 are shown in Fig. 18.35 and the intensity distribution as predicted by Eq. (18.50) for $N = 4$ is shown in Fig. 18.36. When N is very large the principal maxima will be much more intense in comparison to the secondary maxima. We may mention here two points:

(a) A particular principal maximum may be absent if it corresponds to the angle which also determines the minimum of the single-slit diffraction pattern. This will happen when

$$d \sin\theta = m\lambda \qquad \textbf{(18.58)}$$

and

$$b \sin\theta = \lambda, 2\lambda, 3\lambda, \ldots \qquad \textbf{(18.59)}$$

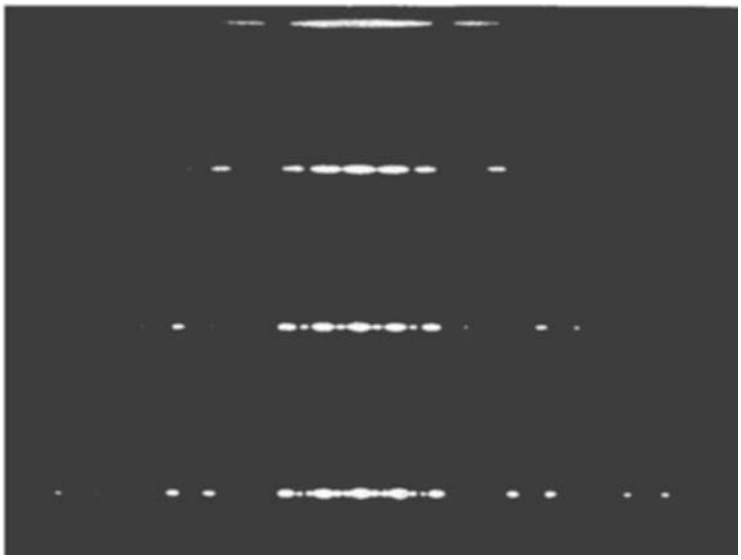


Fig. 18.35 The multiple-slit Fraunhofer diffraction patterns corresponding to b = 0.0044 cm, d = 0.0132 cm and $\lambda = 6.328 \times 10^{-5}$ cm. The number of slits are 1, 2, 3 and 4 respectively (Adapted from Ref. 18.17; used with permission).

are satisfied simultaneously and is usually referred to as a missing order. Even when Eq. (18.59) does not hold exactly (i.e., if $b \sin\theta$ is close to an integral multiple of λ), the intensity of the corresponding principal maximum will be very weak (see, for example, Fig. 18.36 around $\theta \approx 0.8°$).

(b) In addition to the minima predicted by Eq. (18.56), we will also have the diffraction minima (see Eq. 18.55); however, when N is very large, the number of such minima will be very small.

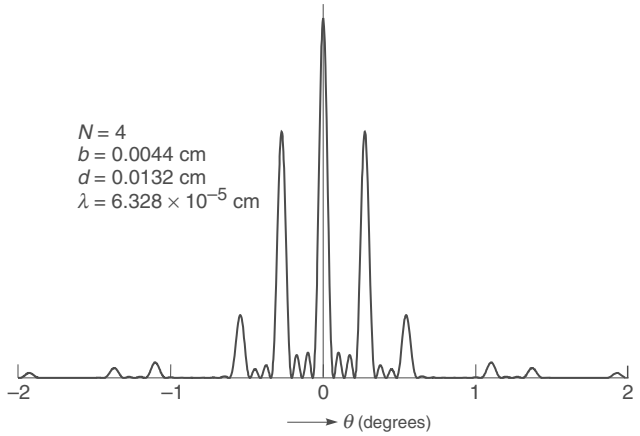


Fig. 18.36 The intensity distribution corresponding to the four-slit Fraunhofer diffraction pattern as predicted by Eq. (18.50) corresponding to b = 0.0044 cm, d = 0.0132 cm and $\lambda = 6.328 \times 10^{-5}$ cm. The principle maxima occur at $\theta \approx 0.275°$, $0.55°$, $0.82°$, $1.1°$, ... Notice the (almost) absent third order.

18.7.2 Width of the Principal Maxima

We have shown above that in the diffraction pattern produced by N slits, the mth order principal maximum occurs at

$$d \sin\theta_m = m\lambda, m = 0, 1, 2, \ldots \quad \textbf{(18.60)}$$

Further, the minima occur at the angles given by Eq. (18.57). If $\theta_m + \Delta\theta_{1m}$ and $\theta_m - \Delta\theta_{2m}$ represent the angles of diffraction corresponding to the first minimum on either side of the principal maximum, then $\frac{1}{2}(\Delta\theta_{1m} + \Delta\theta_{2m})$ is known as the **angular half width of the *m*th order principal maximum**. For a large value of N, $\Delta\theta_{1m} \simeq \Delta\theta_{2m}$ which we write as $\Delta\theta_m$. Clearly,

$$d \sin(\theta_m \pm \Delta\theta_m) = m\lambda \pm \frac{\lambda}{N} \quad \textbf{(18.61)}$$

But

$$\sin(\theta_m \pm \Delta\theta_m) = \sin\theta_m \cos\Delta\theta_m \pm \cos\theta_m \sin\Delta\theta_m$$

$$\simeq \sin\theta_m \pm \Delta\theta_m \cos\theta_m \quad \textbf{(18.62)}$$

Thus Eq. (18.61) gives us

$$\Delta\theta_m \simeq \frac{\lambda}{Nd\cos\theta_m} \quad \textbf{(18.63)}$$

which shows that the principal maximum becomes sharper as N increases.

18.8 THE DIFFRACTION GRATING

LO 7

In Sec. 18.6, we have discussed the diffraction pattern produced by a system of parallel equidistant slits. An arrangement which essentially consists of a large number of equidistant slits is known as a **diffraction grating**; the corresponding diffraction pattern is known as the grating spectrum. Since the exact positions of the principal maxima in the diffraction pattern depend on the wavelength, the principal maxima corresponding to different spectral lines (associated with a source) will correspond to different angles of diffraction. Thus the grating spectrum provides us with an easily obtainable experimental set up for determination of wavelengths. From Eq. (18.63) we see that for narrow principal maxima (i.e., sharper spectral lines), a large value of N is required. A good quality grating, therefore, requires a large number of slits (typically about 15,000 per inch). This is achieved by ruling grooves with a diamond point on an optically transparent sheet of material; the grooves act as opaque spaces. After each groove is ruled, the machine lifts the diamond point and moves the sheet forward for the ruling of the next groove. Since the distance between two consecutive grooves is extremely small, the movement of the sheet is obtained with the help of the rotation of a screw which drives the carriage carrying it. Further, one of the important requirements of a good quality grating is that the lines should be as equally spaced as possible; consequently, the pitch of the screw must be constant, and it was not until the manufacture of a nearly perfect screw (which was achieved by Rowland in 1882) that the problem of construction of gratings was successfully solved. Rowland's arrangement gave 14,438 lines per inch, corresponding to $d = 2.54/14438 = 1.759 \times 10^{-4}$ cm. For such a grating, for $\lambda = 6 \times 10^{-5}$ cm, the maximum value of m would be 2, and, therefore, only the first two orders of the spectrum will be observed. However, for $\lambda = 5 \times 10^{-5}$ cm, the third order spectrum will also be visible.

Commercial gratings are produced by taking the cast of an actual grating on a transparent film like that of cellulose acetate. An appropriate strength solution of cellulose acetate is poured on the ruled surface and allowed to dry to form a strong thin film, detachable from the parent grating. These impressions of a grating are preserved by mounting the film between two glass sheets. Nowadays, gratings are also produced holographically, where one records the interference pattern between two plane or spherical waves (see Example 14.5). In contrast to ruled gratings, holographic gratings have a much larger number of lines/cm.

18.8.1 The Grating Spectrum

In Sec. 18.6, we have shown that the positions of the principal maxima are given by

$$d \sin \theta = m\lambda; \quad m = 0, 1, 2, \ldots \qquad \textbf{(18.64)}$$

This relation, which is also called the **grating equation**, can be used to study the dependence of the angle of diffraction θ on the wavelength λ. The zeroeth order principal maximum occurs at $\theta = 0$ irrespective of the wavelength. Thus, if we are using a polychromatic source (e.g., white light) then the central maximum will be of the same colour as the source itself. However, for $m \neq 0$, the angles of diffraction are different for different wavelengths and, therefore, various spectral components appear at different positions. Thus, by measuring the angles of diffraction for various colors one can (knowing the value of m) determine the values of the wavelengths. It may be mentioned that the intensity is maximum for the zeroeth order spectrum (where no dispersion occurs) and it falls off as the value of m increases.

If we differentiate Eq. (18.64), we would obtain

$$\frac{\Delta\theta}{\Delta\lambda} = \frac{m}{d \cos\theta} \qquad \textbf{(18.65)}$$

From this result we can deduce the following conclusions:

(a) Assuming θ to be very small (i.e., $\cos\theta \simeq 1$) we can see that the angle $\Delta\theta$ is directly proportional to the order of spectrum (m) for a given $\Delta\lambda$, so that for a given m, $\Delta\theta/\Delta\lambda$ is a constant. Such a spectrum is known as a **normal spectrum** and in this the difference in angle for two spectral lines is directly proportional to the difference in wavelengths. However, for large θ, it can easily be shown that the dispersion is greater at the red end of the spectrum.

(b) Equation (18.65) tells us that $\Delta\theta$ is inversely proportional to d, and therefore smaller the grating element, the larger will be the angular dispersion.

Figures 18.37 and 18.38 show schematic diagrams of the experimental arrangement for studying the grating spectrum of a polychromatic source. In Fig. 18.37, we have shown a small hole placed at the focal plane of the lens L_1. A parallel beam of white light emerging from L_1 falls on the grating and the diffraction pattern is observed on the focal plane of the lens L_2. If instead of a hole we have a slit at the focal plane of L_1 (see Fig. 18.38) — as it is indeed the case in a typical laboratory set up — we would have parallel beams propagating in different directions, and in the focal plane of the lens L_2 we will have a band spectrum as shown in Fig. 18.38.

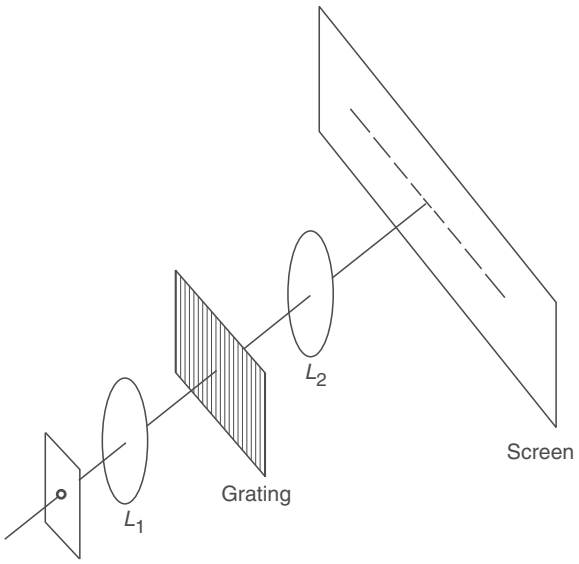


Fig. 18.37 Fraunhofer diffraction of a plane wave incident normally on a grating.

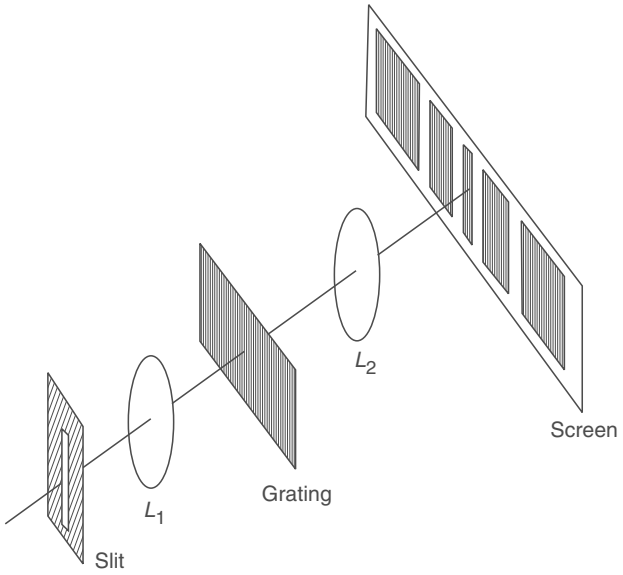


Fig. 18.38 If instead of a point source we have a slit in the focal plane of L_1 then one will obtain bands on the focal plane of L_2.

The lens L_2 is the objective of a telescope and the diffraction pattern is viewed through an eyepiece. The angles of diffraction for various orders of the grating spectrum can be measured and knowing the value of d, one can calculate the wavelength of different spectral lines.

Example 18.10 Consider a diffraction grating with 15,000 lines per inch. **(a)** Show that if we use a white light source the second and third order spectra overlap. **(b)** What will be the angular separation of the D_1 and D_2 lines of sodium in the second order spectra?

Solution: **(a)** The grating element is

$$d = \frac{2.54}{15000} = 1.69 \times 10^{-4} \text{ cm}$$

Let θ_{mv} and θ_{mr} represent the angles of diffraction for the mth order spectrum corresponding to the violet and red colors, respectively. Thus,

$$\theta_{2v} = \sin^{-1} \frac{2 \times 4 \times 10^{-5}}{1.69 \times 10^{-4}} \simeq \sin^{-1} 0.473 \simeq 28.2^\circ$$

$$\theta_{2r} = \sin^{-1} \frac{2 \times 7 \times 10^{-5}}{1.69 \times 10^{-4}} \simeq \sin^{-1} 0.828 \simeq 55.90^\circ$$

and

$$\theta_{3v} = \sin^{-1} \frac{3 \times 4 \times 10^{-5}}{1.69 \times 10^{-4}} \simeq \sin^{-1} 0.710 \simeq 45.23^\circ$$

where we have assumed the wavelengths of the violet and red colours to be 4×10^{-5} cm, and 7×10^{-5} cm, respectively. Since $\theta_{2r} > \theta_{3v}$, the second and third order spectra will overlap. Further since sin $\theta_{3r} > 1$, the third order spectrum for the red colour will not be observed.

(b) Since $d \sin \theta = m\lambda$, we have for small $\Delta\lambda$:

$$(d \cos \theta)\, \Delta\theta = m(\Delta\lambda)$$

or

$$\Delta\theta = \frac{m\Delta\lambda}{d\left\{1 - \left(\frac{m\lambda}{d}\right)^2\right\}^{1/2}}$$

$$\simeq \frac{2 \times 6 \times 10^{-8}}{1.69 \times 10^{-4}\left[1 - \left(\frac{2 \times 6 \times 10^{-5}}{1.69 \times 10^{-4}}\right)^2\right]^{1/2}}$$

$$\simeq 0.0010 \text{ radians} \simeq 3.47'$$

Thus, if we are using telescope of angular magnification 10, the two lines will appear to have an angular separation of 34.7′.

18.8.2 Resolving Power of a Grating

In the case of a grating the resolving power refers to the power of distinguishing two nearby spectral lines and is defined by the following equation:

$$R = \frac{\lambda}{\Delta\lambda} \qquad \textbf{(18.66)}$$

where $\Delta\lambda$ is the separation of two wavelengths which the grating can just resolve; the smaller the value of $\Delta\lambda$, the larger the resolving power.

The Rayleigh criterion (see Sec. 18.4) can again be used to define the limit of resolution. According to this criterion, if the principal maximum corresponding to the wavelength $\lambda + \Delta\lambda$ falls on the first minimum (on the either side of the principal maximum) of the wavelength λ, then the two wavelengths λ and $\lambda + \Delta\lambda$ are said to be just resolved (see Fig. 18.39). If this common diffraction angle is represented by θ and if we are looking at the mth order spectrum, then the two wavelengths λ and $\lambda + \Delta\lambda$ will be just resolved if the following two equations are simultaneously satisfied:

$$d \sin \theta = m(\lambda + \Delta\lambda) \tag{18.67}$$

and

$$d \sin \theta = m\lambda + \frac{\lambda}{N} \tag{18.68}$$

Thus,

$$R = \frac{\lambda}{\Delta\lambda} = mN \tag{18.69}$$

which implies that the resolving power depends on the total number of lines in the grating—obviously on only those lines which are exposed to the incident beam (see the derivation in Sec. 18.6). Further, the resolving power is proportional to the order of the spectrum. Thus, to resolve the D_1 and D_2 lines of sodium ($\Delta\lambda$ = 6 Å) in the first order, N must be at least $(5.89 \times 10^{-5})/(6 \times 10^{-8}) \approx 1{,}000$.

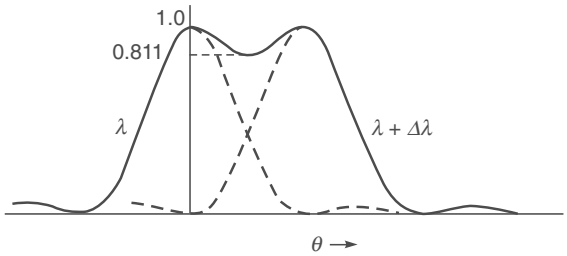


Fig. 18.39 The Rayleigh criterion for the resolution of two spectral lines.

From Eq. (18.69) it appears that the resolving power of the grating would increase indefinitely if N is increased; however, for a given width of the grating $D(= Nd)$, as N is increase, d decreases and therefore the maximum value of m also decreases. Thus if d becomes 2.5λ, only first and second order spectra will be seen and if it is further reduced to about 1.5λ then only the first order spectrum will be seen.

18.8.3 Resolving Power of a Prism

We conclude this section by calculating the resolving power of a prism. Figure 18.40 gives a schematic description of the experimental arrangement for observing the prism spectrum which is determined through the following formula:

$$n(\lambda) = \frac{\sin \dfrac{A + \delta(\lambda)}{2}}{\sin \dfrac{A}{2}} \tag{18.70}$$

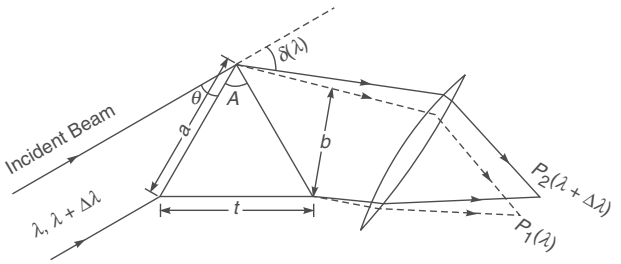


Fig. 18.40 The schematic of the experimental arrangement to observe the prism spectrum. P_1 and P_2 represent the images corresponding to λ and $\lambda + \Delta\lambda$, respectively.

where A represents the angle of the prism and δ the angle of minimum deviation. We assume that the refractive index decreases with λ (which is usually the case) so that δ also decreases with λ. In Fig. 18.40, the points P_1 and P_2 represent the images corresponding to λ and $\lambda + \Delta\lambda$, respectively. We are assuming that $\Delta\lambda$ is small so that the same position of the prism corresponds to the minimum deviation position for both wavelengths. In an actual experiment one usually has a slit source (perpendicular to the plane of the paper) forming line images at P_1 and P_2. Since the faces of the prism are rectangular, the intensity distribution will be similar to that produced by a slit of width b (see Sec. 18.2)*. For the lines to be just resolved the first diffraction minimum [m = 1 in Eq. (18.12)] of λ should fall at the central maximum of $\lambda + \Delta\lambda$, thus we must have

$$\Delta\delta \approx \frac{\lambda}{b} \tag{18.71}$$

In order to express $\Delta\delta$ in terms of $\Delta\lambda$, we differentiate Eq. (18.70):

$$\frac{dn}{d\lambda} = \frac{1}{\sin \dfrac{A}{2}} \cos\left[\frac{A + \delta(\lambda)}{2}\right] \frac{1}{2} \frac{d\delta}{d\lambda}$$

Thus,

$$\Delta\delta = \frac{2 \sin \dfrac{A}{2}}{\cos \dfrac{A + \delta(\lambda)}{2}} \frac{dn}{d\lambda} \Delta\lambda$$

* Since we have a slit source we need not consider the diffraction in a direction perpendicular to the plane of the diagram.